Qualcomm Technologies, Inc.

SW6100 Storage Guide

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hyungchul.won@lge.com

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Contents

1 Storage overview **3**
1.1 eMMC/SDIO use cases . . . 4

2 Storage features **5**
2.1 eMMC device features . . . 5
2.2 SDIO features . . . 6
2.3 Hardware features . . . 6
2.4 PSRAM features . . . 7

3 Interfaces **8**
3.1 eMMC interface . . . 8
3.2 SDIO block diagram . . . 8
3.3 PSRAM block diagram . . . 8

4 Configure storage features **9**
4.1 Enable SDHC using kernel configuration . . . 9
4.2 Modify SDHCI card power management states . . . 9
4.3 Configure device using eMMC parameters in UEFI . . . 10
4.4 Configure eMMC Linux kernel device tree . . . 11
4.5 Configure part-specific PSRAM DTSI parameters . . . 13

5 Debug common storage driver issues **16**
5.1 Debug with sysfs or debugfs nodes . . . 16

6 References **17**
6.1 Related documents . . . 17
6.2 Acronyms and terms . . . 17

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hyungkyu.kim@lge.com

1 Storage overview

The SW6100 storage guide describes the features and supported device types, enabling you to configure, customize, and deploy storage capabilities optimally and efficiently.

The following table lists the storage capabilities of SW6100.

SW6100 storage type features

Slot	Bits	I/O voltage (V)	Card voltage (V)	Maximum frequency	Timing modes	Functions
SDC1	8	1.8	2.95	• SDR – 192 MHz • DDR – 192 MHz	• DS, HS • HS200 • HS400 • DDR	eMMC 5.1
SDC2	4	1.8/2.95	2.95	• SDR – 200 MHz • DDR – 50 MHz	• DS • HS • SDR12 • SDR25 • SDR50 • SDR104 • DDR50	• SDIO 3.0 • SD 3.0

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Slot	Bits	I/O voltage (V)	Card voltage (V)	Maximum frequency	Timing modes	Functions
PSRAM Arm® Cortex®-M55 processor	8/16	1.2	1.98	SDR – 200 MHz	• SDR • DDR	PSRAM

1.1 eMMC/SDIO use cases

•Flash device images

Device firmware or system images can be flashed onto the device using various tools such as Qualcomm Flash Image Loader (QFIL), Joint Test Action Group (JTAG), Fastboot, and Qualcomm Product Support Tools (QPST).

•Boot from eMMC

During the boot process, the device reads essential boot images stored in the eMMC. These include components like the secondary boot loader (SBL), recovery image, kernel, resource power manager (RPM), and TrustZone firmware.

•User data storage

The user data partition in eMMC is managed by the High-Level Operating System (HLOS), where user-generated content and application data are stored.

•Secure data storage

Sensitive data related to TrustZone security is stored in the RPM partition of the eMMC, ensuring secure handling of trusted operations.

•Pseudo-static random access memory (PSRAM)

PSRAM serves as an extended memory module, used for display, audio, OTA updates, and edge AI tasks like quick data processing.

•SDIO functionality

Secure digital input output (SDIO) is used to connect the SD cards to host devices, enabling extra I/O functionalities such as Wi-Fi, Bluetooth®, GPS, or NFC through a standardized SD interface.

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2 Storage features

SW6100 supports the following features for the various storage types.

2.1 eMMC device features

The following eMMC device features are supported.

Feature	SW6100 support	Description
Boot and storage	Yes	Driver support for boot/storage from an 8-bit eMMC card
eMMC RPMB	Yes	Driver support for read/write access to RPMB
eMMC HS200	Yes	• Supports eMMC HS200 bus speed mode • SDR data sampling with frequency up to 200 MHz, data rate up to 200 MBps
eMMC HS400	Yes	• Supports eMMC HS400 bus speed mode • DDR data sampling with frequency up to 200 MHz, data rate up to 400 MBps (from eMMC 5.0)

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Feature	SW6100 support	Description
eMMC 5.1	Yes	eMMC 5.1 supported with backward compatibility, firmware update, and command queuing features, Enhanced Data Strobe, RPMB Write Throughput, Secure Write Protection, Cache Flush Policy/Cache Barrier, AUTO BKOPS

2.2 SDIO features

The following SDIO card features are supported.

Feature	SW6100 support	Description
SDIO for WLAN	Yes	Driver support for SDIO to use with WLAN

2.3 Hardware features

The following hardware features are supported.

Hardware features	SW6100 support	Description
SDCC5* core	2 cores	• SDC1 – 8 bit • SDC2 – 4 bit
Clock speeds supported	Yes	• 75 MHz • 93.75MHz • 166 MHz • 200 MHz • Max CLK rate is 200 MHz
Low-power mode (LPM)	Yes	Only VDD (VCC) goes to 0 V on runtime suspend and system suspend.

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Note:

- SDCC5 is the fifth generation of the SDCC core.
- The internal DLL and pad configuration is different for each SDC slot, so each SDC slot supports different timing modes.

2.4 PSRAM features

Features	Description
Interface type	PSRAM support through hexadecimal/octal-SPI interfaces
Extensible RAM	PSRAM acts as Extensible RAM
Data rate	512 Mb X 8/x16
Clock speeds supported	• 75 MHz • 93.75 MHz • 166 MHz • 200 MHz (max CLK rate is 200 MHz)

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3 Interfaces

Following interfaces are supported in SW6100.

3.1 eMMC interface

The following diagram shows the eMMC part of MCP connect with the SD card.

Note: For VDDI measurement (in μF), see the eMCP vendor recommendation.

Figure1 eMMC connection diagram

3.2 SDIO block diagram

The following diagram shows the various GPIO and PMIC connections for the SD card.

Figure2 SD card connection diagram

3.3 PSRAM block diagram

The following diagram shows the PSRAM block diagram.

Figure3 PSRAM connection diagram

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hyungchul.woo@lge.com

4 Configure storage features

Use the following storage configurations and device tree parameters for UFS and SD card functionalities.

4.1 Enable SDHC using kernel configuration

The following table lists the default values of kernel configuration to enable eMMC and SDIO card functionality.

Configuration	Enabled by default (yes/no)	Description
CONFIG_MMC	Yes	Enables MMC driver
CONFIG_MMC_DEBUG	No	Enables driver debug messages
CONFIG_MMC_SDHCI	Yes	Detects SDHCI driver
CONFIG_MMC_CQHCI	Yes	Enables command queue host control interface
CONFIG_MMC_SDHCI_MSM	Yes	Enables MSM SDHCI driver
CONFIG_MMC_BLOCK	Yes	Enables MMC block driver
CONFIG_MMC_CRYPTO	Yes	Enables support for the MMC crypto engine

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hyungchul.won@lge.com

4.2 Modify SDHCI card power management states

To optimize power management, modify the SDHCI card power management states as required. However, altering the states can cause changes in power consumption.

SDHC card power management states

State	Description
Runtime suspend or resume	• The SDIO enters the runtime suspend state after 50 ms of inactivity. • Resume occurs with the next SD card operation. This state is enabled by default and is not configurable. • The eMMC transitions to the runtime suspend state after 50 milliseconds of inactivity. It resumes with the next I/O operation. This feature is always enabled and cannot be configured.
System suspend or resume	• The SD card enters system suspend state with the suspend event. • SD card system resume state occurs with the system resume event. • This state is enabled by default and is not configurable. • The eMMC enters system suspend mode upon receiving a suspend event and resumes operation with a system resume event. This functionality is always enabled and cannot be configured.

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The following table list the voltage states during the SDHCI suspend states.

Voltage state

Condition	Clock gating	VDD state	VDD-IO state
Runtime suspend	Yes	Off	On
System suspend	Yes	Off	On

4.3 Configure device using eMMC parameters in UEFI

The device tree specifies the SD card parameters such as drive strength, maximum speed mode, and HSR settings. The device tree is at `<workspace_root_path>/core.boot/6.5/Settings/Soc/Aspen/Core/Storage/SDCC/sdcc.dtsi`.

The following is an example of the eMMC device tree node in UEFI.

Device tree node/key	Device tree value	Description
`DriveStrength`	0x1EDB	`DriveStrength` as recommended, sets SDC CLK, CMD, and data drive strengths to 7, 7, 4, respectively.
MaxSpeedMode	SDCC_HS400_MODE	Defines maximum speed mode. For eMMC the value is SDCC_HGS400_MODE.

4.4 Configure eMMC Linux kernel device tree

The SoC DTSI node specifies register address space, clocks, interrupts, and reset information. The platform DTSI node specifies power supply, voltage, and current levels. The `sdhc_1:mmc@4744000` node describes the on-chip SDHC host controller. This node is in the `<workspace_root_path>/sources/kernel/kernel_platform/kernel/arch/arm64/boot/dts/qcom/vienna.dtsi` file.

Property	Description
`compatible`	Specifies the compatible hardware string. For example, “qcom,sdhci-msm-v5” indicates this node uses the Qualcomm MSM SDHCI controller version 5.
`reg`	Provides the physical base addresses and size of memory regions used by the controller. This includes SDHC, CQHCI, and ICE block registers.
`reg-names`	Names associated with each memory region in reg. Typically used by drivers to access specific hardware blocks (“hc”, “cqhci”, “ice”).
`iommus`	Defines the IOMMU device (apps_smmu) and Stream ID (0xC0) used for DMA address translation.
`qcom,iommu-dma`	Platform-specific property to configure DMA behavior. “bypass” disables IOMMU translation.
`interrupts`	Lists IRQs (GIC SPI numbers) and their trigger types (IRQ_TYPE_LEVEL_HIGH) used by the SDHCI hardware (host controller and power IRQ).
`interrupt-names`	Corresponds to the names of interrupts, e.g., “hc_irq” for SDHC and “pwr_irq” for power-related events.

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Property	Description
`clocks`	Specifies the clocks required by the controller: main core, interface, and ICE (inline crypto engine) clocks.
`clock-names`	Names for each of the above clocks: “core”, “iface”, and “ice_core”.
`qcom, ice-clk-rates`	Defines the operational frequencies for the ICE engine, e.g., 300 MHz and 100 MHz.
`interconnects`	Describes interconnect paths using phandles and endpoints for bandwidth voting—used for SDHC-to-DDR and CPU-to-SDHC links.
`interconnect-names`	Names each interconnect path defined above, like “sdhc-ddr” and “cpu-sdhc”.
`operating-points-v2`	Points to an OPP (Operating Performance Points) table (e.g., &sdhc1_opp_table) defining frequency-voltage pairs supported by SDHC.
`qcom, dll-hsr-list`	Specifies DLL (Delay Locked Loop) configuration settings for High Speed Rates (HSR) mode; platform-specific tuning values.
`qcom, restore-after-cx-collapse`	Indicates that hardware state should be restored after collapse of the CX power domain.
`mmc-ddr-1_8v`	Enables DDR mode operation at 1.8V signaling.
`mmc-hs200-1_8v`	Enables HS200 mode operation at 1.8V signaling.
`mmc-hs400-1_8v`	Enables HS400 mode operation at 1.8V signaling.
`mmc-hs400-enhanced-strobe`	Enables enhanced strobe feature in HS400 mode for better data reliability.
`no-sd`	Indicates that this controller will not support SD cards.
`no-sdio`	Indicates that this controller will not support SDIO devices.
`bus-width`	Number of data lines (1, 4, or 8). Here it uses 8-bit data bus.
`non-removable`	Indicates that the device (eMMC) is soldered to the board and not hot-swappable.
`supports-cqe`	Indicates that the controller supports Command Queue Engine (CQE) for better performance in eMMC.

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Property	Description
qcom, devfreq, freq-table	Frequency values (in Hz) used for device frequency scaling (e.g., 50 MHz, 200 MHz).
qcom, scaling-lower-bus-speed-mode	Indicates the fallback mode to use during frequency scaling. Example: “DDR52”.
cap-mmc-hw-reset	Enables support for MMC hardware reset line.
mmc-rst-n-disable	Prevents toggling of the rst_n line; used when it’s shared or not functional.
resets	Phandle to reset controller; GCC_SDCC1_BCR controls reset behavior of the SDCC1 block.
reset-names	Describes the name of the reset line, here it is “core_reset”.
status	Marks the node as “disabled” or “okay” to indicate hardware availability.
qos0 { mask, vote }	Defines QoS settings (e.g., mask = <0x0f>, vote = <43>) for interconnect bandwidth voting.

4.5 Configure part-specific PSRAM DTSI parameters

To configure settings specific to a PSRAM part, see the DTSI file: `<workspace>/dt_settings/soc/aspen_lpai_awm/core/psram/psram_part.dtsi`.

The following are the current DTSI node settings for the **APS256XXN** PSRAM part.

```
&soc {
psram: qcom-psram-controller {
status = "okay";
compatible = "qcom,psram-controller";
#address-cells = <1>;
#size-cells = <1>;
io-voltage = <1200>;
spi-max-frequency = <DT_FREQ_M(200)>; // move it to part
specific
page-size = <DT_SIZE_K(2)>;
size = <DT_SIZE_M(64)>;

aps256xxn: APS256XXN {
  status = "disabled";
  compatible = "apmem,aps256xxn";
  #address-cells = <1>;
  #size-cells = <1>;
```

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```
reg = <0x28000000 0x10000000>;
address-length = <0x4>;

use-burst-op;
burst-opcode = <0x20 0xA0>;
reg-opcode = <0x40 0xC0>;
reset-opcode = <0xFF>;
reset-addr-mask = <0x00 0x00>;

latency-type-addr = <0x00>;
fixed-latency;
reg-latency-cycles = <5 1>;
mem-latency-cycles = <10 5>;

ddr;
bus-width = <8>;
bus-width-addr-mask = <0x08 0x00>;
sleep-addr-mask = <0x06 0xF0>;
dpd-addr-mask = <0x06 0xC0>;
row-boundary-crossing;
rbx-addr = <0x08>;
};
```

You can add or modify part-specific parameters within the corresponding DTSI node.

The following the details of each parameter in the PSRAM memory part **aps256xxn: APS256XXN**.

Parameter	Description
`status = "disabled"`	Indicates the device is currently disabled in the device tree.
`compatible`	Specifies the compatible string for device matching (“apmem,aps256xxn”).
`#address-cells = <1>`	Defines the number of address cells used in the child nodes.
`#size-cells = <1>`	Defines the number of size cells used in the child nodes.
`reg = <0x28000000 0x10000000>`	Base address and size of the memory-mapped region.
`address-length = <0x4>`	Length of the address field in bytes.
`use-burst-op`	Enables burst operation mode.
`burst-opcode = <0x20 0xA0>`	Opcodes used for burst read/write operations.
`reg-opcode = <0x40 0xC0>`	Opcodes used for register read/write operations.

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Parameter	Description
`reset-opcode = <0xFF>`	Opcode used to reset the device.
`reset-addr-mask = <0x00 0x00>`	Address mask used during reset operation.
`latency-type-addr = <0x00>`	Address used to configure latency type.
`fixed-latency`	Indicates that the device uses fixed latency.
`reg-latency-cycles = <5 1>`	Latency cycles for register operations (read/write).
`mem-latency-cycles = <10 5>`	Latency cycles for memory operations (read/write).
`ddr`	Indicates the device supports DDR (Double Data Rate) mode.
`bus-width = <8>`	Width of the data bus in bits.
`bus-width-addr-mask = <0x08 0x00>`	Address mask for bus width configuration.
`sleep-addr-mask = <0x06 0xF0>`	Address mask used to put the device into sleep mode.
`dpd-addr-mask = <0x06 0xC0>`	Address mask used to enter Deep Power Down (DPD) mode.
`row-boundary-crossing`	Indicates support for crossing row boundaries in memory access.
`rbx-addr = <0x08>`	Address used for row boundary crossing configuration.

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5 Debug common storage driver issues

For more information about debugging, see SDHCI Architecture and debugging.

5.1 Debug with sysfs or debugfs nodes

To identify the status or update the state, use the *sysfs* and *debugfs* nodes.

Mount *debugfs* to ensure that the *debugfs* file system is accessible for the debug nodes.

```
mount -t debugfs none /sys/kernel/debug
```

The following table lists the eMMC device state commands.

Description	Command
eMMC error stats information	`cat /sys/kernel/debug/mmc0/err_stats`
eMMC latest error state details	`cat /sys/kernel/debug/mmc0/err_state`
Current running clock details of an eMMC	`cat sys/kernel/debug/mmc0/ios`
eMMC runtime status information	`cat /sys/devices/platform/soc@0/8804000.mmc/power/runtime_status`
eMMC auto suspend delay time	`cat /sys/devices/platform/soc@0/8804000.mmc/power/autosuspend_delay_ms`
eMMC runtime active time	`cat /sys/devices/platform/soc@0/8804000.mmc/power/runtime_active_time`
eMMC runtime suspend time	`cat /sys/devices/platform/soc@0/8804000.mmc/power/runtime_suspended_time`
eMMC uevent details	`cat /sys/devices/platform/soc@0/8804000.mmc/uevent`

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6 References

6.1 Related documents

Title	Resource
Qualcomm Technologies, Inc.	
SDHCI Architecture and Debugging	80-NL199-1
Standards	
EXT4	https://opensource.com/article/17/5/introduction-ext4-filesystem

6.2 Acronyms and terms

Acronym or term	Definition
AH8	Auto-hibern8
DTSI	Device tree source inclusion
GPIO	General purpose input output
JEDEC	Joint electron device engineering council
LUN	Logical unit number
PVL	Preferred vendor list
PSRAM	Pseudo-static random access memory
RPMB	Replay protected memory block
SD	Secure digital
SDHCI	Secure digital host controller interface
SSD	Solid state drive
WLUN	Well-known logical unit number

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